

Infestations

1. Scabies

2. Lice

Scabies

Scabies (*Sarcoptes scabiei*):

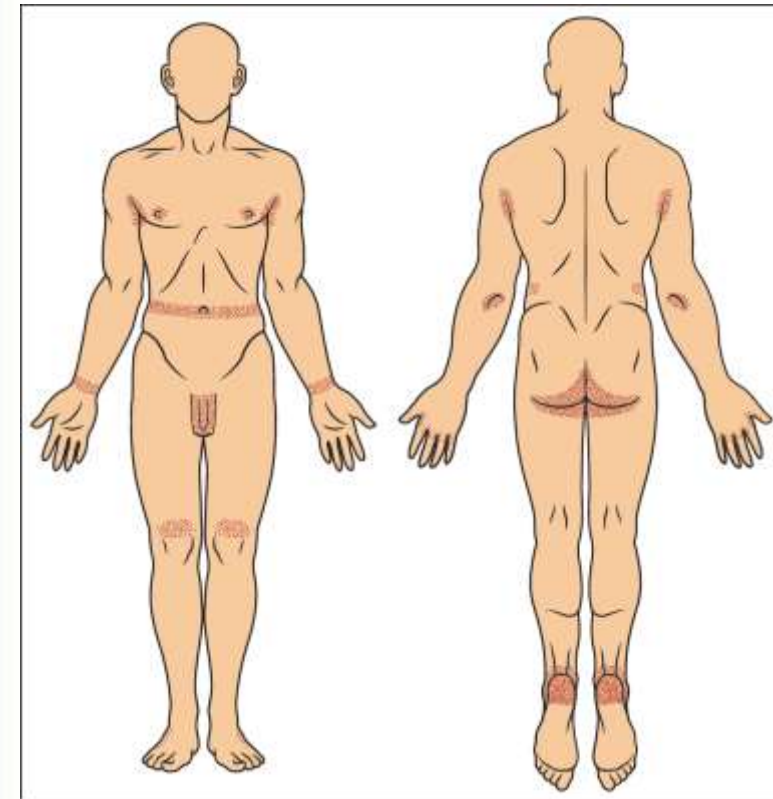
- The most common infestation worldwide.
- Causes intense itching that characteristically keeps those affected awake at night.
- The female mite burrows into the epidermis and lays eggs.

Scabies:

Box 17.3 Scabies – points to note

- There may be very few burrows, though the patient has widespread itching.
- The distribution of the infestation is characteristically the fingers, wrists, nipples, abdomen, genitalia, buttocks and ankles.
- Close personal contact is required for infestation to occur – for example, within a family, through infants in playgroups, and through regular nursing of elderly patients.
- Itching may persist even after all mites have been eliminated; itching papules on the scrotum and penis are particularly persistent.

Sparing face (in adults) and back



Transmission:

- Transmission occurs because of **close personal contact** (at least 15 min of skin-to-skin contact) with an infected individual.
- The first symptoms of itching occur **2 weeks** later when the immune system reacts to the proteins in the mites, eggs, and feces in the skin.

Diagnosis:

History Taking:

- Several individuals in the same household/institution/ classroom/ward are being affected simultaneously by a rash that is intensely itchy at night.
- Severe itching, **more at night, classic distribution.**

Physical Examination:

1. Burrows:

- Clinically, burrows can be seen, especially in the finger-web spaces and on the genitals
- Burrows are linear palpable ridges on the skin with a black speck indicating the position of the mite, which can be teased out of its burrow using a sterile needle and mounted onto a microscope slide.

2. Patients often have a widespread **papular rash**, which is due to a reaction to the infestation, with multiple **excoriation marks** which can become secondarily infected with staphylococcus.

Scabies burrows:



**Linear palpable ridges on the skin with a black speck
indicating the position of the mite**

Scabies in children:

- Babies and young children infested with scabies characteristically present with **erythematous cutaneous papules** and **nodules** in the **axillae** and on the **soles of the feet**.
- Lesions might blister.
- Classic **burrows are rarely seen** in this age group.

Scabies in an infant:



Scabies nodules in a child



Crusted scabies:

- Crusted scabies can look similar to **dry scaly skin rashes** such as psoriasis and eczema, and consequently can be misdiagnosed.
- Patients are usually **immunosuppressed** or **elderly** and do **not complain of itching**, as their immune cells are not reacting against the mite proteins. Consequently, **mite numbers are usually in the hundreds.**



Crusted scabies clinical findings:

- Crusted **fine superficial scaling on the skin.**
- **Very little erythema** (unlike psoriasis and eczema)
- If the diagnosis is missed then many close contacts such as nurses and caregivers develop classic scabies, and Small outbreaks can occur.



Crusted scabies:



Management:

- A full explanation of how to use topical therapy is essential. (m.c.c of treatment failure is incorrect use)
- The lotions should be applied from the **neck downwards**. (although the head and neck of babies should also be treated)
- Patients should be told that they and all their close personal contacts need to be treated at the same time. (to prevent reinfestation)
- Towels, bedding, and underwear should be washed.

Management:

- The treatment should be left **overnight** and then repeated after 7 days. (i.e. **D1 and D8 applications**)
- They should pay particular attention to the web spaces and genital areas.
- They should reapply the lotion after washing their hands.

Management:

- Patients should be warned that the skin itching will take 6–8 weeks to subside.
- Persistent itching often leads patients to conclude that the mites are still active and they subsequently treat themselves repeatedly, leading to an **irritant dermatitis (Post scabies dermatitis)**
- The itching resolves when the mites, eggs, and feces have been removed from the skin by the host's immune cells.

Management:

1. **Permethrin cream 5%:**

- First-line treatment
- Should be left overnight, two applications 7 days apart, adults apply from the neck downwards; babies/infants apply to all the skin.

2. **Malathion lotion 0.5%**

- Second line
- Applied as permethrin.

3. **Ivermectin:**

- Dose: 200 µg/kg
- Two doses 7 days apart
- Can be given to immunocompromised patients and those with crusted scabies (avoid in patients < 15 kg and in pregnancy).

Management:

- If permethrin and malathion are not available, then >> 10% **sulphur** in yellow soft paraffin is effective and safe.
- 25% **benzyl benzoate** emulsion may also be used.
- For severe pruritus >>> Topical anti pruritic agents e.g. crotamiton.

Lice

- Head Lice
- Body Lice
- Pubic Lice

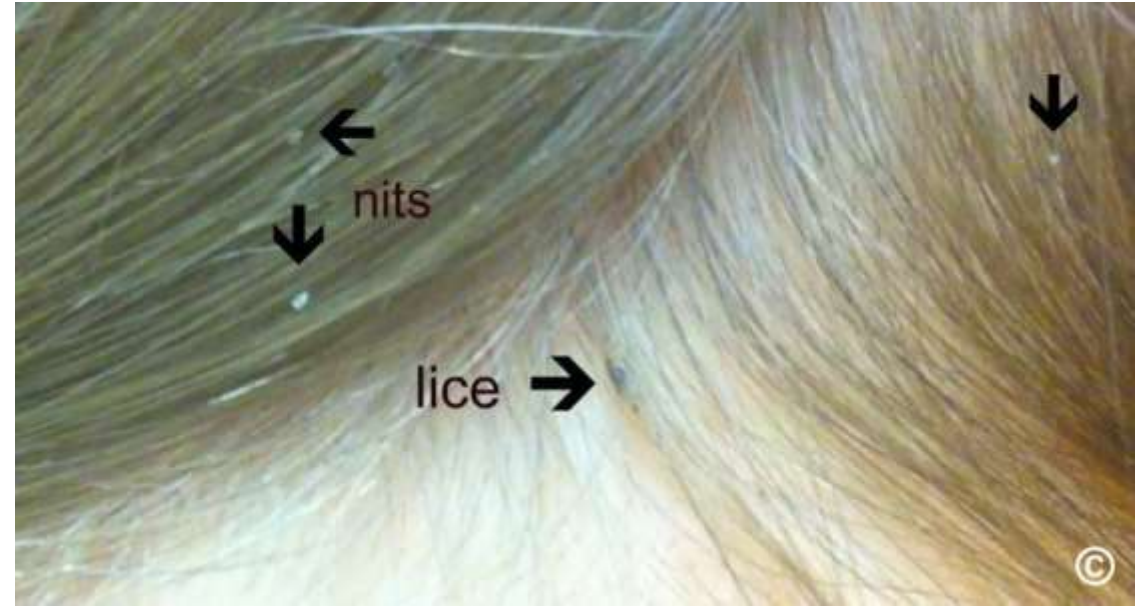
Head lice:

- They have a worldwide distribution and can affect anyone.
- Children are the most common hosts.
- Lice are transmitted by **head-to-head contact**, and on **combs, brushes, and hats**
- Girls are more commonly affected than boys



Head lice:

- **Mild itching** may be the only symptom of head lice.
- Inspection of the hair close to the scalp may reveal **adult lice and nits** (white empty egg cases) in infested individuals.
- Patients may develop an **itchy irritant-looking dermatitis** on their upper back and neck area.
- **Fine-toothed combs** can aid detection.



Management:

- Only treat individuals with live lice visible on the scalp.
- **Fine-toothed combs** can be used to comb out lice and eggs.
- An application of hair conditioner usually allows the comb to pass more easily through the hair.
- Combinations of **insecticides** and **Fine-toothed combs (bug-busting)** are effective.



Management:

- First-line treatment is **permethrin 1–5% crème** rinse applied to dry hair and left on overnight.
- Alternative agents include **phenothrin** 0.5% or **malathion** 0.5%

Body lice:

- Body lice are the vectors of several human pathogens including:
 - **Bartonella Quintana** (agent of trench fever, bacillary angiomatosis, and endocarditis).
 - **Rickettsia Prowazekii** (agent of typhus).
- Body lice tend to affect individuals from poorer economic backgrounds

Body lice:

- The lice live in the **host's clothes** and bite the skin.
- Close inspection, especially of the **clothing seams**, reveals the adults and eggs.
- Infested individuals have a widespread papular eruption with excoriations.



Body lice:

Infested individuals have a **widespread papular eruption with excoriations.**



Management

- Wash clothing in hot water OR place them in a sealed plastic bag for **2 weeks** to kill adults and eggs.
- Treat the skin reactions with a **moderately potent topical steroid**, plus **topical antibiotic** if secondarily infected with bacteria.
- Take blood for culture and serology if louse-borne systemic disease is suspected. (e.g. if the patient has a fever or constitutional symptoms)

Pubic lice

- These lice prefer the sparser hair-bearing sites on the skin such as the **pubic**, **axillary**, and **eyelash** area.
- The so-called **crab lice** are slow-moving and are spread by **close personal contact**.
- Check the patient for other **sexually transmitted diseases**.



Pubic lice on eyelashes.



Pubic lice on the Pubic area

Management:

- Use topical **permethrin** 5% cream or 0.5% **malathion** to the skin from the neck downwards, left on overnight, repeated after 7 days.
- If the eyelashes are involved use **petrolatum only**, as insecticides can damage the eyes.

Thank You